



Graduate Program Update to External Advisory Board

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GRADUATE PROGRAM OVERVIEW

Demographics and Recruiting:

- 54 Graduate Students:
 - 44 PhD (36% domestic: up from 20% last year)
 - 10 MS (80% domestic)
 - 54% Male and 46% Female
- International students primarily from India, Iran (paused), and China (decreasing).
- Increasing numbers from Nigeria.
- Recruiting concerns:
 - How do we recruit effectively from other countries?
 - Many international students cannot afford graduate school in US (MS degree).
 - **How do we increase MS enrollment?**



MS CHEMICAL ENGINEERING

MS Degree Options:

- Four formats (36 credit hours):
 - Two-year
 - One-year fast track
 - 4+1 BS/MS
 - **Co-op track:** Professional Skills courses and one semester industry co-op experience (coursework-only)
- All students gain advanced chemical engineering knowledge with customizable coursework and options for thesis, project, or coursework-only paths.
- Core curriculum covers transport phenomena, thermodynamics, separation processes, reaction engineering, and mathematical methods.
- We are looking for ways to grow our MS programs



INDUSTRY- INTEGRATED MS

The Opportunity:

- Create a Chemical Engineering MS pathway designed for practicing engineers
- Combine advanced technical training with immediate workplace application
- Build the program with industry rather than simply marketing it to industry
- Leverage UIC's location and existing MS infrastructure

Question for the External Advisory Board:

What would an MS program need to provide for your company to enroll employees, sponsor capstone (research) projects, and actively participate?



THE OPPORTUNITY

Who is the program for?

1. Working engineers who want deeper technical capability without leaving employment
2. Employers seeking workforce development tied to real technical challenges
3. Engineers who want an applied MS degree: not necessarily a research thesis or an MBA

Core value proposition

TECHNICAL DEPTH

Advanced ChE
coursework

REAL PROBLEMS

Employer capstone
projects

PROFESSIONAL GROWTH

Business + leadership
coursework

Established models:

<u>Feature</u>	<u>Purdue</u>	<u>UIUC</u>	<u>Stanford Honors Coop</u>
Degree	MS ChE	M.Eng. ChE Leadership	MS ChE
Total credits	30	34	45 quarter units
Format	On-campus / part-time	On-campus / online	Part-time / online
Technical ChE	15 credits	12-credit core + electives	8 graduate ChE courses
Business / leadership	9 credits	14 credits	Optional electives
Industry capstone	6 credits; required	Capstone / practicum option	None required
Employer project	Yes	Yes	Not required
Working professionals	3–4 yr pathway	Full- or part-time	2–5 yr pathway
Primary emphasis	Industry practice	Technical + leadership	Technical depth + flexibility

**UIC opportunity: technical rigor + employer integration +
Chicago access**



UIC FOUNDATION

Build on what UIC already has:

- UIC Chemical Engineering already offers a 36-credit MS
- Existing thesis, project, and coursework-only pathways provide flexibility
- The project pathway provides a natural foundation for an industry capstone

Potential additions for an industry-focused pathway

- Working-professional scheduling and hybrid delivery
- Employer-defined projects with industry co-mentors
- Business, project-management, and leadership electives
- Formal framework for industry participation, IP, and confidentiality



WHY UIC? WHY CHICAGO?

A strong regional value proposition:

UIC sits within a dense and diverse industrial ecosystem.

- Chemicals & materials
- Pharma & biotech
- Food & consumer products
- Water & environment
- Energy & electrification
- Manufacturing & process systems
- Engineering & consulting
- National laboratories

UIC advantage

Employees stay in the region and on the job while coursework and capstone projects strengthen UIC-industry relationships.



PROPOSED CURRICULUM

Strawman curriculum:

For discussion — final structure should be shaped with industry.

ADVANCED ChE (20-24 credits)

- Transport; Thermodynamics; Reactions; Separations; Modeling/Data; Technical electives

PROFESSIONAL (4-8 credits)

- Project management; Economics; Leadership; Communication

CAPSTONE (4–8 credits)

- Employer problem
- Faculty + industry mentor
- Actionable deliverable

**Potential delivery: part-time • evening/hybrid • 2–4 year
completion window**



INDUSTRY CAPSTONE

The centerpiece: a real company problem

- 1 COMPANY** Defines a meaningful technical problem
- 2 STUDENT** Analysis • design • experiments • modeling
- 3 UIC FACULTY** Ensures rigor and academic standards
- 4 INDUSTRY MENTOR** Provides context, data, constraints
- 5 DELIVERABLE** Report + presentation + recommendation

Examples: optimization • separations • decarbonization • water • electrochemistry • data/AI • scale-up • TEA



VALUE TO INDUSTRY

What's in it for employers?

DEVELOP TALENT

Advance existing engineers without requiring them to leave employment.

SOLVE PROBLEMS

Apply coursework and capstones to real technical challenges.

RECRUIT & RETAIN

Create a visible development pathway and stronger access to UIC talent.

BUILD PARTNERSHIPS

Capstones can seed internships, recruiting, sponsored research, and continuing education.

Success metric: companies enroll employees and return with additional projects.



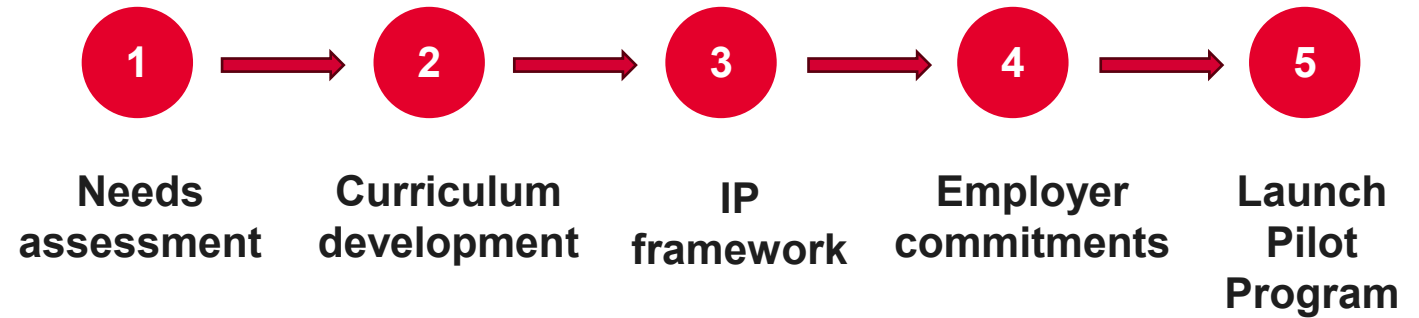
QUESTIONS FOR THE EAB

What do we need to learn from you?

- 1 Would your company support employees pursuing this degree part-time?
- 2 What technical areas would create the most value?
- 3 Would you provide capstone problems and industry mentors?
- 4 How important are evening, hybrid, or online options?
- 5 What would differentiate this from an MBA or conventional MS?
- 6 What barriers (IP, confidentiality, tuition, employee time) must we solve?

NEXT STEPS

Co-design the program with industry:



ASK OF THE EXTERNAL ADVISORY BOARD

Form a working group of EAB members and industry partners to assess demand and help shape the program.

Goal: educate practicing engineers while solving meaningful problems for Chicago-area industry.